

SVD :-

decomposes a matrix A (of size $m \times n$) into three matrices :

$$A = U \Sigma V^T$$

U $m \times m$ orthogonal Matrix (left Singular Vector)

Σ $m \times n$ diagonal matrix of singular values

V^T $n \times n$ orthogonal Matrix (Right Singular Vector)

PROPERTIES :

Columns of U & V are orthonormals

Singular Values in Σ are non negative & arranged in descending order :-

$$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r, 0, \dots, 0)$$

where r is rank of A & $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$

Example :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

1. Compute AA^T & $A^T A$

$$A^T A = \begin{bmatrix} 35 & 44 \\ 44 & 56 \end{bmatrix} \quad AA^T = \begin{bmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{bmatrix}$$

2. Find eigen values & eigenvectors of $A^T A$ & AA^T

Eigen values of $A^T A$: $\lambda_1 = 90.67$ $\lambda_2 = 0.33$

Singular values : $\sigma_1 = \sqrt{90.67} \approx 9.52$

$\sigma_2 = \sqrt{0.33} \approx 0.57$



8. Construct U , Σ , & V :

$$U = \begin{bmatrix} -0.23 & -0.88 \\ -0.52 & -0.24 \\ -0.82 & 0.40 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 9.52 & 0 \\ 0 & 0.57 \\ 0 & 0 \end{bmatrix}$$

$$V = \begin{bmatrix} -0.62 & -0.78 \\ -0.78 & 0.62 \end{bmatrix}$$

1. Verify $A = U\Sigma V^T$

Reduced SVD :-

is compact version of SVD that retains only the non zero singular values & their corresponding singular vectors. If A has rank r , then reduced SVD is :

$$A = U_r \Sigma_r V_r^T$$

U_r

$m \times r$ matrix (first r columns of U)

Σ_r

$r \times r$ diagonal Matrix (non zero singular values)

V_r^T

$r \times n$ matrix (first r rows of V^T)

Doing Same Ex

$$U_r = \begin{bmatrix} -0.23 & -0.88 \\ -0.52 & -0.24 \\ -0.82 & 0.40 \end{bmatrix}$$

$$\Sigma_r = \begin{bmatrix} 9.52 & 0 \\ 0 & 0.57 \end{bmatrix}$$

$$V_r = \begin{bmatrix} -0.62 & -0.78 \\ -0.78 & 0.62 \end{bmatrix}$$

Can Verify

$$A = U_r \Sigma_r V_r^T$$

Eigen Value Decomposition

to factorise a square matrix A (of size $n \times n$) into:

$$A = Q \Lambda Q^{-1}$$

where : Q is an $n \times n$ matrix of eigenvectors

Λ is an $n \times n$ diagonal matrix of eigenvalues

Properties :

- It's only possible for diagonalizable matrices.
- The eigenvectors in Q are linearly independent
- Eigenvalues in Λ represent the scaling factors along the eigenvectors.

Ex:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

1. Eigen Value

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = \lambda^2 - 7\lambda + 10 = 0$$

$$\lambda_1 = 5 \quad \lambda_2 = 2$$

2. Eigen Vector

$$\lambda_1 = 5 \quad (A - 5I)v_1 = 0 \Rightarrow \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 0$$

$$\underline{v_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda_2 = 2$

$$(A - 2I)v_2 = 0 \Rightarrow \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = 0$$

$$\underline{v_2} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

3. Construct Q & Λ

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

4. Verify $A = Q \Lambda Q^{-1}$

$$U \Sigma V^T$$

PROPER STEP BY STEP SVD :-

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} \quad (1) \text{ Compute } A^T A \text{ \& } A A^T$$

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}_{2 \times 3}$$

$$A A^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 & 1 \cdot 3 + 2 \cdot 4 & 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 1 + 2 \cdot 3 & 3 \cdot 3 + 2 \cdot 4 & 3 \cdot 5 + 2 \cdot 6 \\ 5 \cdot 1 + 2 \cdot 5 & 5 \cdot 3 + 2 \cdot 6 & 5 \cdot 5 + 2 \cdot 6 \end{bmatrix}_{3 \times 3}$$

$$= \begin{bmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} 1 \cdot 1 + 3 \cdot 3 + 5 \cdot 5 & 1 \cdot 2 + 3 \cdot 4 + 5 \cdot 6 \\ 2 \cdot 1 + 4 \cdot 3 + 6 \cdot 5 & 2 \cdot 2 + 4 \cdot 4 + 6 \cdot 6 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 35 & 44 \\ 44 & 56 \end{bmatrix}$$

Step (2) Eigen Values of $A^T A$

$$\det(A^T A - \lambda I) = 0$$

$$\det \begin{bmatrix} 35 - \lambda & 44 \\ 44 & 56 - \lambda \end{bmatrix} = (35 - \lambda)(56 - \lambda) - 44^2 = 0$$

$$\lambda^2 - 91\lambda + 24 = 0$$

$$\lambda_1 = 90.67 \quad \lambda_2 \approx 0.33$$

Eigen Vector of $A^T A$:

for $\lambda_1 \approx 90.67$

$$(A^T A - \lambda_1 I) v_1 = 0 \Rightarrow \begin{bmatrix} 35 - 90.67 & 44 \\ 44 & 56 - 90.67 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 0$$

$$v_1 \approx \begin{bmatrix} 0.62 \\ 0.78 \end{bmatrix}$$

for λ_2 :

$$\begin{bmatrix} 35 - 0.33 & 44 \\ 44 & 56 - 0.33 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = 0$$

$$v_2 \approx \begin{bmatrix} -0.78 \\ 0.62 \end{bmatrix}$$

3. Eigen value & Vector of $A A^T$ are same just additional eigenvalue of 0. $\therefore A A^T$ is symmetric. So Eigen vectors of $A A^T$ can be computed. Now $U \Sigma V$

4. 1. Singular values as the square roots of eigen v.
 $\sigma_1 = \sqrt{90.67} \approx 9.52$
 $\sigma_2 = \sqrt{0.33} \approx 0.57$

Construct $\Sigma \approx \begin{bmatrix} 9.52 & 0 \\ 0 & 0.57 \\ 0 & 0 \end{bmatrix}$

Matrix V :

Columns of V are eigen vectors of $A^T A$

$$V = \begin{bmatrix} 0.62 & -0.78 \\ 0.78 & 0.62 \end{bmatrix} \checkmark$$

Matrix U : are eigen vector of $A A^T$

$$U = \begin{bmatrix} -0.23 & -0.88 \\ -0.52 & -0.24 \\ -0.82 & 0.40 \end{bmatrix}$$

§ Verify $A = U \Sigma V^T$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

for Eigen decomposition

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

SVD

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

Eigen Vector

$$(A - \lambda I)$$

$$A \begin{bmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{bmatrix} = 0$$

$$(1-\lambda)(1-\lambda) - 4 = 0$$

$$1 + \lambda^2 - 2\lambda - 4 = 0$$

$$(1-\lambda)^2 = 4$$

$$\lambda = 3, -1$$

$$(1-\lambda) = \pm 2$$

for $\lambda = 3$ ($AV = \lambda V$)

$$\begin{pmatrix} 1-3 & 2 \\ 2 & 1-3 \end{pmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (A - \lambda I) \begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix}$$

$$v_1 \quad \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2v_{11} + 2v_{12} = 0$$

$$2v_{11} - 2v_{12} = 0$$

$$\lambda = 1$$

$$v_2$$

$$\begin{pmatrix} 1-1 & 2 \\ 2 & 1-1 \end{pmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda_1 = 3 \quad \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2x + 2y = 0 \Rightarrow x = y$$

$$2x - 2y = 0 \Rightarrow x = y$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1$$

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x + 2y = 0$$

$$2x$$

$$x = -y$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Q^{-1} = \frac{1}{2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \quad Q^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

SVD $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\underline{AA^T} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}_{3 \times 2}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2} \quad \text{Reduces}$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0 \quad (2-\lambda)^2 = +1$$

$$2-\lambda = 1 \quad 2-\lambda = -1$$

$$\lambda = 1 \quad \lambda = 3$$

$$\lambda_1 = 3$$

$$v_1 \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad v_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow x + y = 0 \quad x = -y$$



$$\lambda_0 = 1 \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + y = 0$$

$$x = -y$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$V^T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (3 \times 2)$$

$$U \Sigma V^T$$

$$AA^T = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_1 = \underline{\underline{3, 1, 0}}$$

$$\lambda_1 = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x + y + z = 0$$

$$x - 2y = 0$$

$$x = 2y$$

$$x - 2z = 0$$

$$x = 2z$$

$$y = 1 \quad \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} y \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + y + z = 0$$

$$x = 0$$

$$y = -z$$

$$\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda_3 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$2x + y + z = 0$$

$$x + y = 0$$

$$x + z = 0$$

$$x = -y$$

$$x = -z$$

$$\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

PCA :-

$$\begin{array}{c|c|c} 1 & 2 & 3 \\ 0 & 4 & 5 \\ -1 & 7 & 0 \\ -1 & 7 & 6 \\ 8 & 4 & 10 \\ \hline 1 & 2 & 3 \end{array}$$

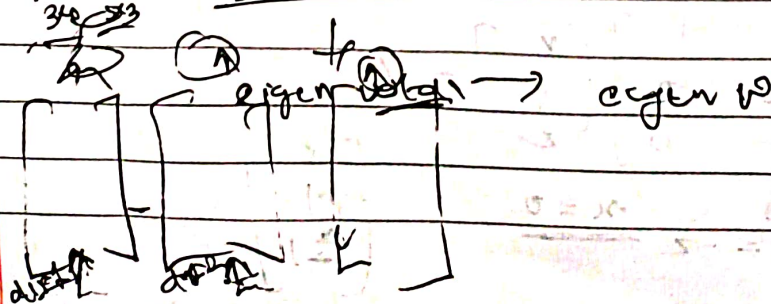
3x7

$A (3 \times 3)$

$A^T = 3 \times 5$

dimension x dimension

$A^T A = 3 \times 3$ Covariance



Q1

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \text{scalar} \quad (2)$$

projecto

$$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (9)$$

$$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 + 2.5 \\ 2 + 2.5 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 4.5 \end{bmatrix}$$

PCA is dimensionality reduction technique used in ML to transform a high dimensional dataset into a lower-dimensional space while preserving as much variance as possible.

Step 1. Standardize the Data

∵ PCA is affected by different scales of features we first standardize the data (mean=0, variances=1)

Ex Dataset :-

X =	2	3
	3	4
	4	5
	5	6
	6	7

Step 1. Compute Mean

$$\mu_1 = \frac{2+3+4+5+6}{5} = 4$$

$$\mu_2 = \frac{3+4+5+6+7}{5} = 5$$

Step 2. Subtract the Mean (Center the data)

$$X_{\text{centered}} = X - \mu = \begin{bmatrix} 2-4 & 3-5 \\ 3-4 & 4-5 \\ 4-4 & 5-5 \\ 5-4 & 6-5 \\ 6-4 & 7-5 \end{bmatrix}$$

$$X_{\text{centered}} = \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Step 3. Compute Covariance Matrix

$$C = \frac{1}{n-1} X_{\text{centered}}^T X_{\text{centered}}$$

$$X_{\text{centered}}^T = \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix}$$

$$X_{\text{centered}}^T X_{\text{centered}} = \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

divide by $n-1 = 5-1 = 4$

$$C = \begin{bmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \end{bmatrix}$$

Compute Eigen Value Vector

$$V = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

Transform Data to Principal Component

Project Data onto eigenvectors

$$X_{PCA} = X_{centered} \cdot V$$

$$X_{PCA} = \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

$$X_{PCA} = \begin{bmatrix} -2.828 & 0 \\ -1.414 & 0 \\ 0 & 0 \\ 1.414 & 0 \\ 2.828 & 0 \end{bmatrix} \xrightarrow{\text{further reduce}} \begin{bmatrix} -2.828 \\ -1.414 \\ 0 \\ 1.414 \\ 2.828 \end{bmatrix}$$

Quadratic form:

Let $K \in \mathbb{R}^{n \times n}$

$$q(x) = x^T A x$$

where x is a n -D vector

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 4 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Positive Semidefinite

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is said to be PSD

if for all $x \in \mathbb{R}^n$

we have $q(x) = x^T A x \geq 0$

Positive Definite (PD)

$$q(x) = x^T A x > 0$$

PSD $\rightarrow$ Eigen values are non negative (≥ 0)

PD $\rightarrow$ " " " " > 0